

READING PASSAGE 3

The Expanding Universe and Hubble's Law

A In recent years, the expanding universe and Hubble's law has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C The mechanisms underlying these phenomena operate at multiple levels of analysis, from the micro-level of individual decision-making to the macro-level of institutional and structural change. At the individual level, cognitive processes, emotional responses, and social influences shape the choices that people make. At the organisational level, formal rules, informal norms, and power dynamics govern collective action. At the societal level, historical legacies, cultural values, and political structures create the broad context within which lower-level processes unfold.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E Despite the substantial progress that has been made, significant challenges and controversies remain. Critics have raised important methodological concerns regarding the generalisability of findings derived from studies conducted in specific contexts. The predominance of research carried out in Western, educated, industrialised, rich, and democratic societies raises questions about the applicability of current theories to populations with different cultural, economic, and political circumstances. Furthermore, the reliance on cross-sectional designs in much of the existing literature limits the capacity to draw strong causal inferences.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G In conclusion, the study of the expanding universe and Hubble's law represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.

Questions 3–4

Choose the correct letter, A, B, C or D.

- 3 Which of the following best describes the author's view on future developments?
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations
- 4 According to the passage, what is one of the key factors identified by researchers?
 - A. too much data collection
 - B. the generalisability of findings across contexts
 - C. insufficient government funding
 - D. overly complex statistical methods

Question 5

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 5 Case studies have documented the effects of _____ change on the system.

Questions 6–7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 6 The passage suggests that structural factors are less important than individual choices.
- 7 Policy-makers have fully implemented all research recommendations.

Questions 8–10

Choose the correct letter, A, B, C or D.

- 8 The meta-analyses cited in the passage reveal that:
 - A. economic incentives and institutional constraints
 - B. individual personality traits

- C. random chance
- D. government policies alone

9 According to the passage, what is one of the key factors identified by researchers?

- A. excessive funding
- B. theoretical and methodological limitations
- C. public interest
- D. international collaboration

10 According to the passage, what is one of the key factors identified by researchers?

- A. too much data collection
- B. the generalisability of findings across contexts
- C. insufficient government funding
- D. overly complex statistical methods

Question 11

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

11 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 12–13

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

12 The landmark study discussed in the passage involved participants from multiple countries.

13 All critics agree that the research findings are universally applicable.